Examiner only

13. *(a)* Explain what is meant by a carbon-neutral fuel. [3]

....

....

....

....

(b) Ethane and an unknown alkane were burned in oxygen.

It follows from the equation below that one volume of ethane produced two volumes of carbon dioxide and three volumes of water vapour.

$$C_2H_6(g) \quad + \quad \tfrac{7}{2}O_2(g) \quad \longrightarrow \quad 2CO_2(g) \quad + \quad 3H_2O(g)$$

10 cm^3 of the unknown alkane C_xH_y burned according to the following equation.

$$C_xH_y(g) \quad + \quad (x + \tfrac{y}{4})O_2(g) \quad \longrightarrow \quad xCO_2(g) \quad + \quad \tfrac{y}{2}H_2O(g)$$

The total volume of carbon dioxide and water vapour produced was 20 cm^3 more than the original volume of C_xH_y and oxygen. All volumes were measured at the same temperature and pressure.

(i) State the volumes of carbon dioxide and water vapour produced on burning 10 cm^3 of the unknown alkane in terms of x and y. [1]

Volume carbon dioxide = ... cm^3

Volume water vapour = ... cm^3

(ii) Calculate the value of y. [2]

y = ...

6

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

GCE AS/A LEVEL

2410U20-1

S19-2410U20-1

CHEMISTRY – AS unit 2
Energy, Rate and Chemistry of Carbon Compounds

THURSDAY, 23 MAY 2019 – MORNING

1 hour 30 minutes

For Examiner's use only			
	Question	Maximum Mark	Mark Awarded
Section A	1. to 6.	10	
Section B	7.	18	
	8.	15	
	9.	12	
	10.	11	
	11.	14	
	Total	80	

2410U201
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.
Section B Answer **all** questions in the spaces provided.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.10***(a)*.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

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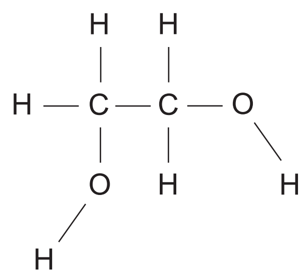
13. (a) In cold weather the wings of an aeroplane can become covered in ice. For safety reasons the wings must be de-iced.

The liquid ethane-1,2-diol, CH_2OHCH_2OH, is used, mixed with water, since this lowers the freezing temperature of water and causes the ice to melt.

(i) Use the diagram below to show the intermolecular forces that allow ethane-1,2-diol to dissolve in water. [3]



(ii) Suggest a reason why the addition of ethane-1,2-diol lowers the freezing temperature of water. [1]

....

....

Examiner only

(b) Ethane-1,2-diol can be oxidised to ethanedioic acid, $(COOH)_2$.

(i) Suggest an oxidising agent suitable to carry out this reaction. [1]

....

(ii) Write the equation for this reaction. Use [O] for the oxidising agent. [2]

....

(iii) To ensure complete oxidation the reagents are refluxed.

Draw and label the apparatus as it is being used in this reaction. [2]

(iv) A sample of the reacting mixture was taken during the reflux process and a mass spectrum was produced. One of the peaks recorded was at m/z 58.

Suggest the identity of the **molecular ion** that gave this peak. [1]

....

Examiner only

(v) The reaction can be used to prepare a sample of solid ethanedioic acid. This is generally hydrated as $(COOH)_2.xH_2O$ where x is an integer.

2.00 g of ethane-1,2-diol was oxidised and 3.94 g of hydrated ethanedioic acid was produced.

Calculate the relative molecular mass of hydrated ethanedioic acid and hence the value of x in its formula. [4]

x = ...

Examiner only

(vi) I. Complete the equation for the reaction which occurs when ethanedioic acid is heated with excess methanol in acidic conditions.

Clearly show the structure of the organic product. [2]

$$\begin{matrix} COOH \\ | \\ COOH \end{matrix} \quad + \quad 2CH_3OH \longrightarrow \qquad\qquad\qquad + $$

II. Name the type of functional group present in the organic product. [1]

....

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END OF PAPER